

## Experiment Name: Modeling a Use Case Diagram

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### Case Study: Career Guidance and Examination Support Portal

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#### Introduction:

This case study focuses on developing an online platform designed to guide students through competitive examination preparation and career counselling. The portal connects students, counsellors, advertisers, and test maintenance authorities under the supervision of an administrator. The system also integrates with external services such as an email server and a secure payment gateway.

Students can register, take mock exams, receive performance feedback, and consult certified counsellors. Counsellors respond to student queries via the portal. Meanwhile, advertisers promote educational content and services through approved advertisements, and test maintenance authorities contribute question papers for practice tests. The administrator controls registrations, approvals, payments, and content management, ensuring the smooth operation of the system.

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#### Objective:

To design a UML Use Case Diagram that visually represents the interaction between multiple users (actors) and the Career Guidance Portal, highlighting how system functionalities are accessed and managed.

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#### Definition:

A Use Case Diagram is a behavioral model within the Unified Modeling Language (UML) that depicts a system's external interactions. It identifies actors (users or external systems) and use cases (functional activities), and represents their communication through associations.

The diagram often includes relationships such as *include* (shared functionality) and *extend* (optional or conditional behavior), which simplify complex processes into clear, modular components.

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### Actors and Use Cases

#### 1. Student

- Register on the portal
- Request test paper
- Answer and submit paper within time
- View evaluation and feedback
- Communicate with counsellors for guidance

- Handle issues like invalid credentials or incomplete forms

## **2. Counsellor**

- Register for counsellor access
- Log in to the system
- Review student queries
- Send counselling replies and updates

## **3. Advertiser**

- Apply for registration as an advertiser
- Log in and choose advertisement schemes
- Submit advertisement content for administrator approval
- Pay advertisement fees via online transaction
- Track advertisement status or upload schedule

## **4. Test Maintenance Authority (TMA)**

- Register and log in to the system
- Upload new test paper details
- Wait for administrator approval
- Receive confirmation and payment upon approval

## **5. Administrator**

- Approve or reject user registrations
- Process test paper upload requests
- Manage advertisement approval and upload
- Dispatch access keys to students
- Handle counsellor and advertiser registration requests
- Maintain records and send notifications through email
- Manage exceptions such as invalid entries, connection failures, and payment issues

## **6. External Systems**

- **Email System:** Sends notifications, activation keys, and responses
  - **Payment Gateway:** Manages secure online transactions
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### Use Case Relationships (based on Diagram):

- Several use cases, such as “Register Student”, “Request Test Paper”, and “Make Request to Counsellor”, include smaller sub-tasks like *Handle Invalid Email*, *Handle Login Failure*, and *Handle Connection Failure*.
  - The Administrator interacts with almost every actor through *include* relationships like *Update User Record* and *Process Upload Requests*.
  - The Advertiser and TMA both share approval-based processes that rely on the administrator’s consent before further action.
  - These include relationships make the diagram modular and prevent duplication of shared activities like *Handle Incomplete Form* or *Handle Invalid Credit Card Number*.
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### Use Case Diagram:

The Use Case Diagram for the Career Guidance and Examination Support Portal (as shown in the attached figure) depicts interactions among all main actors—Student, Counsellor, Administrator, Advertiser, and Test Maintenance Authority—alongside the Email and Payment subsystems.

It clearly presents dependencies through <<include>> relationships, showing how each actor’s operations depend on shared and system-wide actions handled by the administrator.

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### Conclusion:

The developed Use Case Diagram provides a detailed and structured understanding of how various users interact with the Career Guidance Portal. It helps visualize not just the primary actions but also exception handling scenarios (like login or payment failures).

This diagram is a fundamental step in system design, serving as the foundation for advanced UML models such as Sequence Diagrams, Activity Diagrams, and Class Diagrams.

Through this experiment, the overall system behavior and responsibilities of each actor become clear, ensuring better system planning and requirement analysis.